



New material can easily recover valuable material in Li-ion batteries for reuse

Lawrence Berkeley National Laboratory (Berkeley Lab) scientists have invented a new battery material called the Quick-Release Binder, that makes it simple and affordable to separate the valuable material in Li-ion batteries from



Chen Fang holds a vial (left) containing a current collector copper foil from a battery that has been completely freed of valuable electrode components after the Quick-Release Binder dissolved, and another vial (right) where the binder is actively dissolving in alkaline water (Credit: Marilyn Sargent/Berkeley Lab)

the other components and recover them for reuse in a new battery. A battery made with Quick-Release Binder simply needs to be opened, placed in room temperature alkaline water, and gently shaken. The separated elements are easily filtered out of the water and air-dried. Polyacrylic acid (PAA) and polyethylenimine (PEI) are joined through a bond between positively charged nitrogen atoms in PEI and negatively charged oxygen atoms in PAA. When the solid binder material is placed in alkaline water containing sodium hydroxide ($\text{Na} + \text{OH}^-$), the sodium ion pops into the bond site, breaking the two polymers apart. The separated polymers dissolve into the liquid, freeing any electrode components embedded within.

Neuro-robotic bionic arm restores natural arm movement

A team of researchers from Cleveland Clinic have now developed a neurobotic prosthetic arm for patients with upper-limb amputations that allows wearers to think, behave, and function more like a person without an amputation. The investigators developed the bionic prosthetic system to combine three important functions: Intuitive motor control, touch, grip kinesthesia or the intuitive feeling of opening and closing the hand. This bionic system enables wearers to move their prosthetic arm more intuitively and feel sensations of touch and movement at the same time. The



Brain Machine Interface based prosthetic bionic arm (Credit: Cleveland Clinic)

closed-loop neural-machine interface connects with the wearer's amputated limb nerves. It enables patients to send nerve impulses from the brain to the prosthetic limb when they wish to use or move it, as well as to receive physical information from the environment and relay it back to the brain through the nervous system.

Japan rolls out 'humble and lovable' delivery robot

Japanese firm ZMP has introduced RokuRO, a four-wheeled robot that walks the streets of Tokyo delivering groceries. Proponents hope that this robot can address the shortage of delivery workers. The robot won't be operating entirely alone, with humans monitoring remotely and able to intervene. The revised traffic laws will allow self-driving delivery robots to navigate streets across Japan. Regulations set a maximum speed of six kilometres per hour (four



A four-wheeled robot dodges pedestrians on a street outside Tokyo, part of an experiment businesses hope will tackle labour shortages and rural isolation (Credit: <https://techxplore.com>)

miles per hour), meaning the chances of severe injury in the event of a collision are relatively small. Panasonic says its "Hakobo" robot can judge autonomously when to turn as well as detect obstacles, such as construction and approaching bikes, and stop. Test runs so far have ranged from delivering medicine and food to Fujisawa residents to peddling snacks in Tokyo with disarming patterns such as: "Another cold day, isn't it? How about some hot drinks?"